

PAPER_684: UQFF Primordial Black Hole Evaporation: Suppressed Rate and Extended Lifetime

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Framework: Unified Quantum Field Framework (UQFF) v5.30

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Class: UQFFPrimordialBHEvaporation | CP4 #268

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Domain: Primordial Black Holes / Cosmology

Abstract

Primordial black holes evaporate via Hawking radiation with standard lifetime $\tau_{\text{std}}(M) = 5120\pi G^2 M^3 / (\hbar c^4)$. In UQFF the evaporation rate is suppressed by the condensate factor: $\dot{M}_{\text{UQFF}} = \dot{M}_{\text{std}}(1 - f_{\text{TRZ}})(\rho_{\text{SCm}}/\rho_{\text{UA}})e^{-U_m/k_B T_H}$, extending the PBH lifetime by a factor $\tau_{\text{UQFF}}/\tau_{\text{std}} \approx \rho_{\text{UA}}/((1 - f_{\text{TRZ}})\rho_{\text{SCm}}) \approx 11$.

Primary UQFF Equation

$$\dot{M}_{\text{UQFF}} = -\frac{\hbar c^4}{15360\pi G^2 M^2}(1 - f_{\text{TRZ}})\frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}}e^{-U_m/k_B T_H}$$

Parameters

Parameter	Value	Description
M_0	10^{12} kg	Initial PBH mass
$\tau_{\text{UQFF}}/\tau_{\text{std}}$	~ 11	Lifetime extension
t_{form}	10^{-23} s	Formation epoch

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor

Constant	Value	Description
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "UQFFPrimordialBHEvaporation.h"
```

```
UQFFPrimordialBHEvaporation obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFPrimordialBHEvaporationCalculator

calc = UQFFPrimordialBHEvaporationCalculator()
result = calc.compute() # returns all UQFF quantities
print(result)
```

References

Hawking 1974; Carr et al. 2016 (PBH review); Green et al. 2021; UQFF Session 173

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